

Project-A

Blood Pressure Reduction Study Report

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Abstract

This report presents a statistical analysis of a dose-finding study investigating the effect of different concentrations of arginine on blood pressure reduction in hypertensive patients. The study includes 40 patients, randomized into four groups receiving 0, 2, 5, or 10 mg of arginine per day. The primary outcome measured is the systolic blood pressure reduction after two months. Various statistical analyses, including descriptive statistics, hypothesis testing, and linear regression modeling, are conducted to assess the relationship between dosage and blood pressure reduction.

• Introduction

Hypertension is a major risk factor for cardiovascular diseases, and finding effective treatments is crucial. This study evaluates the impact of different doses of arginine on systolic blood pressure reduction. Our objectives include analyzing the distribution of the data, testing statistical hypotheses, and modeling the relationship between dose and blood pressure reduction.

• Methods

.1. Dataset Description

- **Participants:** 40 hypertensive patients (systolic blood pressure ≥ 150 mmHg)
- **Treatment Groups:** 0, 2, 5, and 10 mg/day of arginine
- **Variables:**
 - **dose:** Arginine dosage (mg/day)
 - **bp.reduction:** Systolic blood pressure reduction (mmHg)
 - **gender:** Categorical variable (Male/Female)

.2. Statistical Methods

- **Descriptive statistics:** Mean, median, quartiles, min-max values
- **Graphical analysis:** Histograms, bar plots, scatterplots, and boxplots
- **Outlier detection:** Boxplots
- **Normality and homoscedasticity testing:** Shapiro-Wilk test, Levene's test, Barlett test, F-test, Histogram, QQ-plot, boxplot
- **Statistical inference:** Confidence intervals for blood pressure reduction
- **Hypothesis testing:** two-sample t-test, two-sample welch t-test ANOVA, and tukey post-hoc
- **Regression modeling:** Linear regression model

• R Code and Its Interpretation

.1. Set working directory and load necessary libraries

```
# Set working directory
setwd("/Users/Hossam/Desktop/NU Master's/1st Semester/Statistical Analysis and Visualization/Project")

# Load necessary Libraries
library(ggstatsplot)
library(dunn.test)
library(report)
library(car)
library(multcomp)
library(ggplot2)
library(skimr)
library(wesanderson)
library(RColorBrewer)
```

.2. Data loading and reading

```
# Load data
load("BloodPressure.RData")
```

```
# Check the data structure
str(BloodPressure)
```

Output:

```
## 'data.frame': 40 obs. of 3 variables:
## $ dose : num 0 0 0 0 0 0 0 0 0 0 ...
## $ bp.reduction: num 3 5 -2 -5 0 5 -1 0 -8 -6 ...
## $ gender : int 0 0 0 1 0 0 1 1 1 1 ...
```

Comment: The dataset consists of 40 observations with three variables: dose, blood pressure reduction, and gender.

```
# Skim the data
skim(BloodPressure)
```

Output:

Data summary

Name	BloodPressure
------	---------------

Number of rows 40

Number of columns 3

Column type frequency:

numeric 3

Group variables None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
dose	0	1	4.25	3.81	0	1.50	3.5	6.25	10	
bp.reduction	0	1	7.62	7.93	-8	2.75	6.5	13.00	26	
gender	0	1	0.45	0.50	0	0.00	0.0	1.00	1	

.3. Descriptive Statistics

- Get data summary

```
summary(BloodPressure)
```

Output:

```
##      dose      bp.reduction      gender
##  Min.   : 0.00   Min.   : -8.000   Min.   : 0.00
##  1st Qu.: 1.50   1st Qu.:  2.750   1st Qu.: 0.00
##  Median : 3.50   Median :  6.500   Median : 0.00
##  Mean   : 4.25   Mean   :  7.625   Mean   : 0.45
##  3rd Qu.: 6.25   3rd Qu.: 13.000   3rd Qu.: 1.00
##  Max.   :10.00   Max.   : 26.000   Max.   : 1.00
```

- Calculate the following: mean, median, minimum, maximum, first and third quartile, using separate commands

```
# Calculate mean for Dose
```

```
mean(BloodPressure$dose, na.rm=TRUE)
```

Output:

```
## [1] 4.25
```

```
# Calculate median for Dose
```

```
median(BloodPressure$dose, na.rm=TRUE)
```

Output:

```
## [1] 3.5
```

```
# Calculate minimum value for Dose  
min(BloodPressure$dose, na.rm=TRUE)
```

Output:

```
## [1] 0
```

```
# Calculate maximum value for Dose  
max(BloodPressure$dose, na.rm=TRUE)
```

Output:

```
## [1] 10
```

```
# Calculate quartiles for Dose  
quantile(BloodPressure$dose, na.rm=TRUE, c(0.25, 0.75))
```

Output:

```
## 25% 75%  
## 1.50 6.25
```

```
# Calculate mean for bp.reduction  
mean(BloodPressure$bp.reduction, na.rm=TRUE)
```

Output:

```
## [1] 7.625
```

```
# Calculate median for bp.reduction  
median(BloodPressure$bp.reduction, na.rm=TRUE)
```

Output:

```
## [1] 6.5
```

```
# Calculate minimum for bp.reduction  
min(BloodPressure$bp.reduction, na.rm=TRUE)
```

Output:

```
## [1] -8
```

```
# Calculate maximum for bp.reduction  
max(BloodPressure$bp.reduction , na.rm=TRUE)
```

Output:

```
## [1] 26
```

```
# Calculate quartiles for bp.reduction  
quantile(BloodPressure$bp.reduction , na.rm=TRUE, c(0.25, 0.75))
```

Output:

```
##      25%      75%  
##  2.75 13.00
```

- **Convert Gender variable from numeric into categorical**

```
BloodPressure$gender <- factor(BloodPressure$gender, labels = c("Male", "Female"))
```

- **Frequency table for categorical variables (Gender)**

```
table(BloodPressure$gender)
```

Output:

```
##  
##      Male Female  
##       22      18
```

Comment: There are slightly more male participants (22) than female participants (18).

- **Calculate the correlation coefficient between bp-reduction and Dose**

```
cor(BloodPressure$bp.reduction, BloodPressure$dose, use="complete.obs")
```

Output:

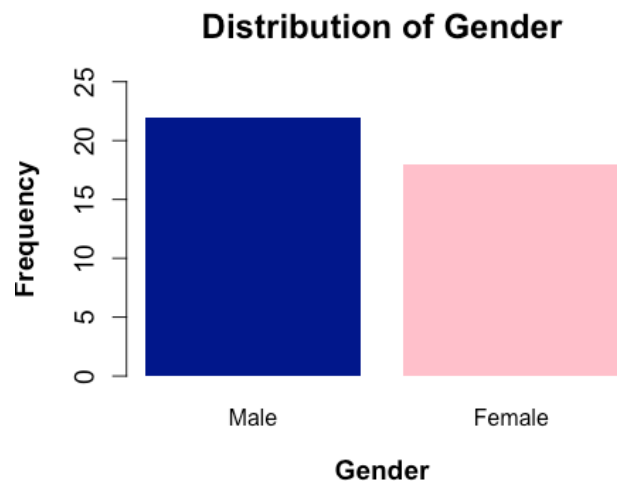
```
## [1] 0.8594944
```

Comment: There is a strong positive correlation (0.8594) between dose and blood pressure reduction.

.4. Graphics

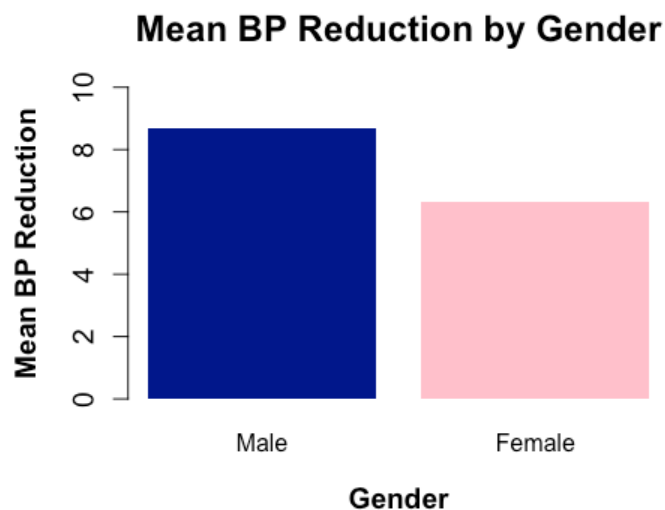
- Generate a bar chart of a categorical variable for the gender

```
barplot(table(BloodPressure$gender), xlab = expression(bold("Gender")), ylab = expression(bold("Frequency")), ylim = c(0,25), col = c("darkblue", "pink"), border=NA, main = "Distribution of Gender", cex.main = 1.5, cex.lab = 1.2, cex.axis = 1.2)
```



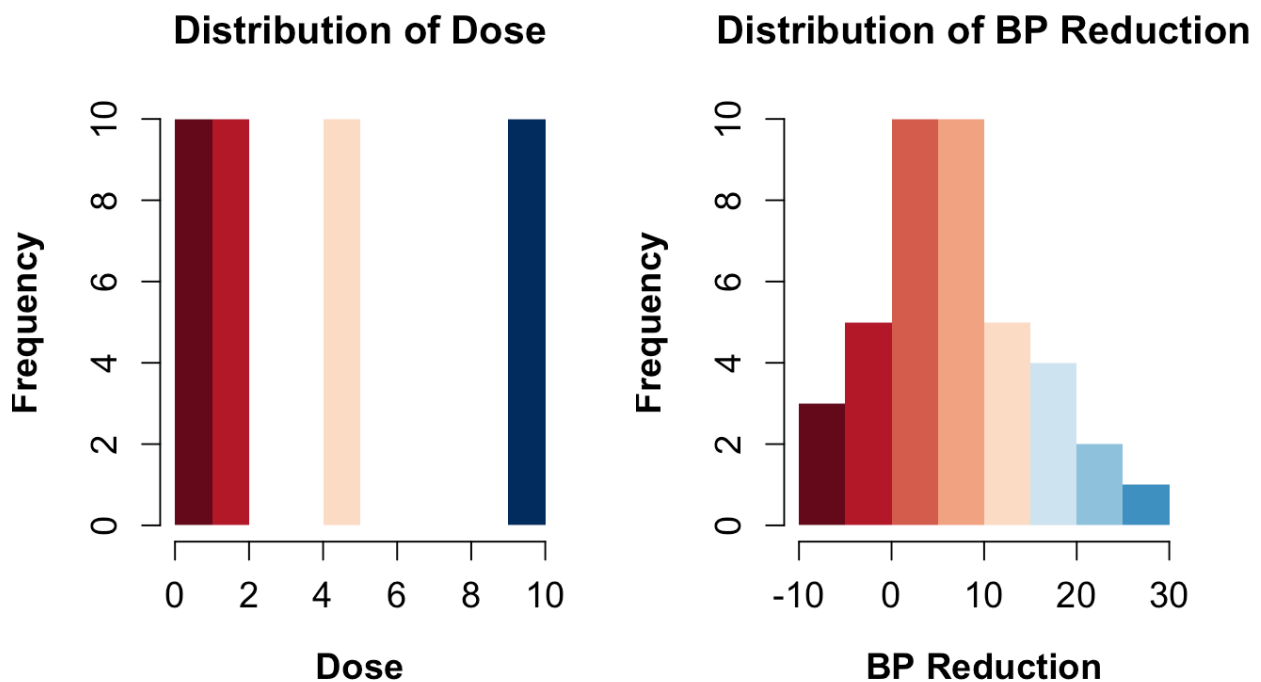
- Generate a bar chart graph with mean bp.reduction in males and females

```
barplot(tapply(BloodPressure$bp.reduction, list(Gender = BloodPressure$gender), mean, na.rm = TRUE), xlab = expression(bold("Gender")), ylab = expression(bold("Mean BP Reduction")), col = c("darkblue", "pink"), border=NA, ylim = c(0,10), main = "Mean BP Reduction by Gender", cex.main = 1.5, cex.lab = 1.2, cex.axis = 1.2)
```



- Make a histogram of a continuous variable “Dose”, “bp.reduction”

```
par(mfrow=c(1,2))
hist(BloodPressure$dose, xlab = expression(bold("Dose")), ylab = expression(bold("Frequency")), main = "Distribution of Dose", col = brewer.pal(n = 10, name = "RdBu"), breaks = 10, border=NA, cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.2)
hist(BloodPressure$bp.reduction, xlab = expression(bold("BP Reduction")), ylab = expression(bold("Frequency")), main = "Distribution of BP Reduction", col = brewer.pal(n = 10, name = "RdBu"), border=NA, cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.2)
```



- Make a scatterplot of 2 continuous variables Dose and bp.reduction, and add the regression lines for each gender

```
par(mfrow=c(1,1))
# Plot scatterplot
plot(BloodPressure$dose, BloodPressure$bp.reduction, xlab = expression(bold("Dose (mg/day)")), ylab = expression(bold("BP Reduction (mmHg)")), main = "Scatterplot of Dose vs BP Reduction by Gender", col = "lightgray", pch = 16, cex = 1.2, cex.axis = 1.2, cex.lab = 1.2, las = 1, bty = "l")

# Points for Males
points(BloodPressure$dose[BloodPressure$gender == "Male"], BloodPressure$bp.reduction[BloodPressure$gender == "Male"], col = rgb(0, 0, 1, 0.6), pch = 16, cex = 1.5)

# Points for Females
```

```
points(BloodPressure$dose[BloodPressure$gender == "Female"], BloodPressure$bp
.reduction[BloodPressure$gender == "Female"], col = rgb(1, 0.4, 0.6, 0.6), pc
h = 16, cex = 1.5)
```

Add Legend

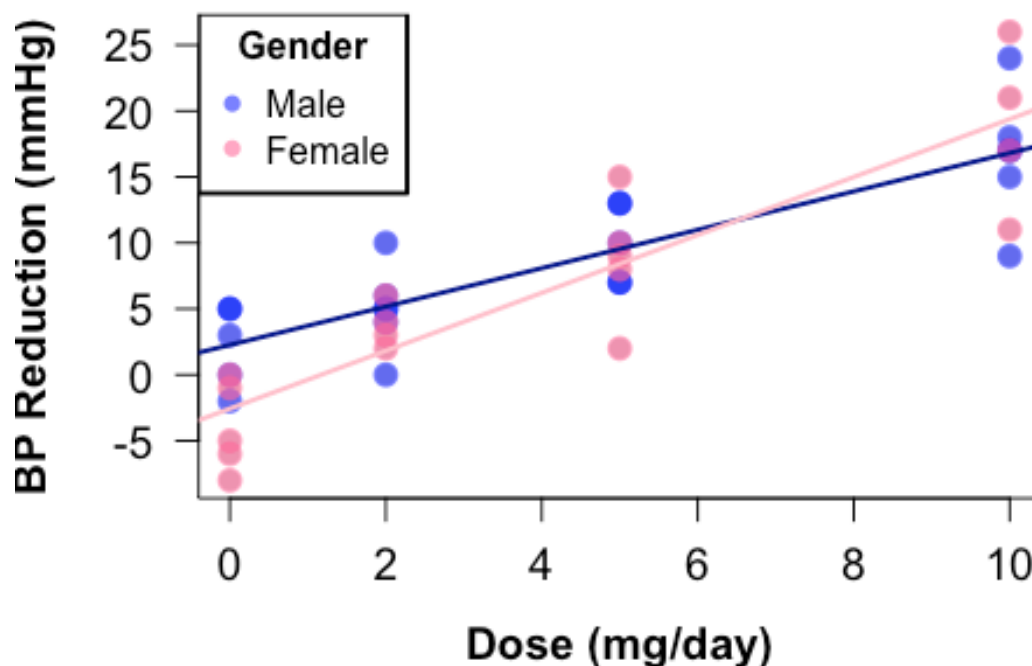
```
legend("topleft", legend = c("Male", "Female"), col = c(rgb(0, 0, 1, 0.6), rg
b(1, 0.4, 0.6, 0.6)), pch = 16, title = expression(bold("Gender")), cex = 1,
y.intersp = 1, box.lwd = 2)
```

Add regression lines

```
abline(lm(BloodPressure$bp.reduction[BloodPressure$gender == "Male"]~BloodPre
ssure$dose[BloodPressure$gender == "Male"]), col="darkblue", lwd = 2)
```

```
abline(lm(BloodPressure$bp.reduction[BloodPressure$gender == "Female"]~BloodP
ressure$dose[BloodPressure$gender == "Female"]), col="pink", lwd = 2)
```

Scatterplot of Dose vs BP Reduction by Gender

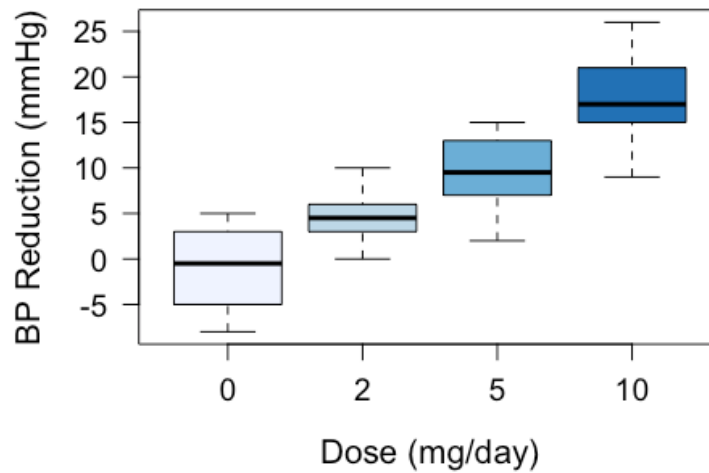


- Make a boxplot of Gender and a separate boxplots per Doses (as.factors)

Boxplot of BP Reduction per Dose

```
boxplot(bp.reduction~dose,data=BloodPressure,main="Boxplot of BP Reduction pe
r Dose", xlab = "Dose (mg/day)", ylab = "BP Reduction (mmHg)", col = brewer.p
al(n = 4, name = "Blues"), border = "black", cex.axis = 1.2, cex.lab = 1.3, c
ex.main = 1.5, cex.sub = 1.2, las = 1)
```

Boxplot of BP Reduction per Dose



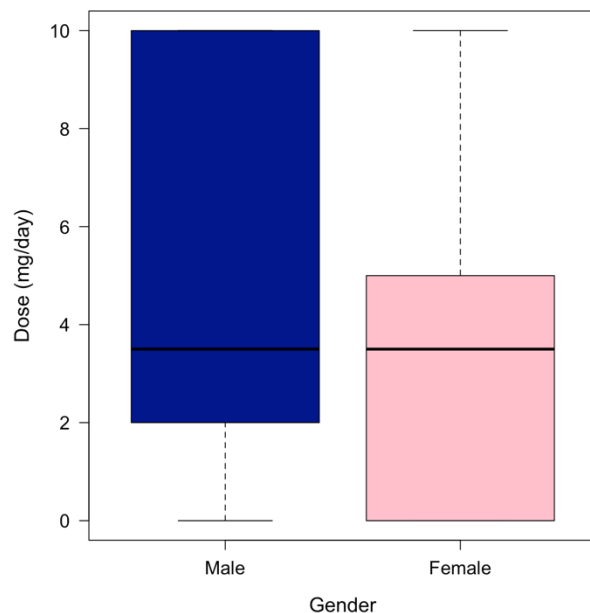
Boxplot Dose per Gender

```
boxplot(dose~gender,data=BloodPressure,main="Boxplot of Gender per Dose", xlab = "Gender", ylab = "Dose (mg/day)", col = c("darkblue","pink"), border = "black", cex.axis = 1.2, cex.lab = 1.3, cex.main = 1.5, cex.sub = 1.2, las = 1)
```

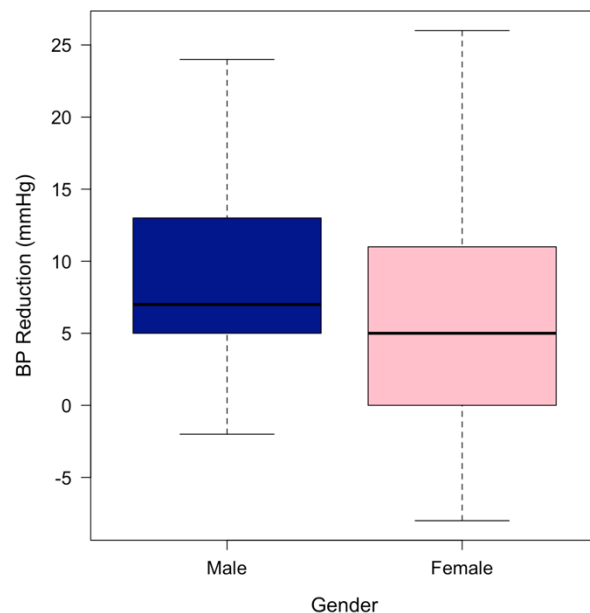
Boxplot BP Reduction per Gender

```
boxplot(bp.reduction~gender,data=BloodPressure,main="Boxplot of BP Reduction per Gender", xlab = "Gender", ylab = "BP Reduction (mmHg)", col = c("darkblue","pink"), border = "black", cex.axis = 1.2, cex.lab = 1.3, cex.main = 1.5, cex.sub = 1.2, las = 1)
```

Boxplot of Gender per Dose



Boxplot of BP Reduction per Gender



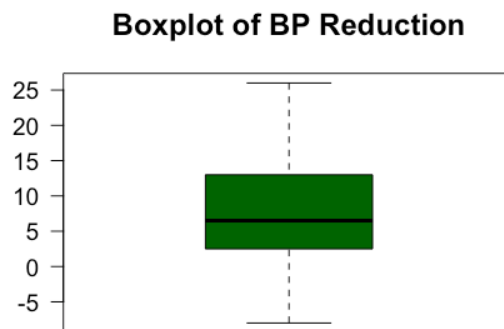
.5. Outlier detection

- Explore the data for any existing outliers, identify them (do NOT remove them if found)

```
outliers<-boxplot(BloodPressure$bp.reduction,plot=TRUE, main="Boxplot of BP R  
eduction", col = "darkgreen", border = "black", cex.axis = 1.2, cex.lab = 1.3  
, cex.main = 1.5, cex.sub = 1.2, las = 1)$out
```

outliers

Output:



```
## numeric(0)
```

Comment: No outliers detected in blood pressure reduction values.

.6. Testing for normality/ homoscedasticity

.6.1. Normality testing:

- Shapiro-Wilk test

```
shapiro.test(BloodPressure$bp.reduction)
```

Output:

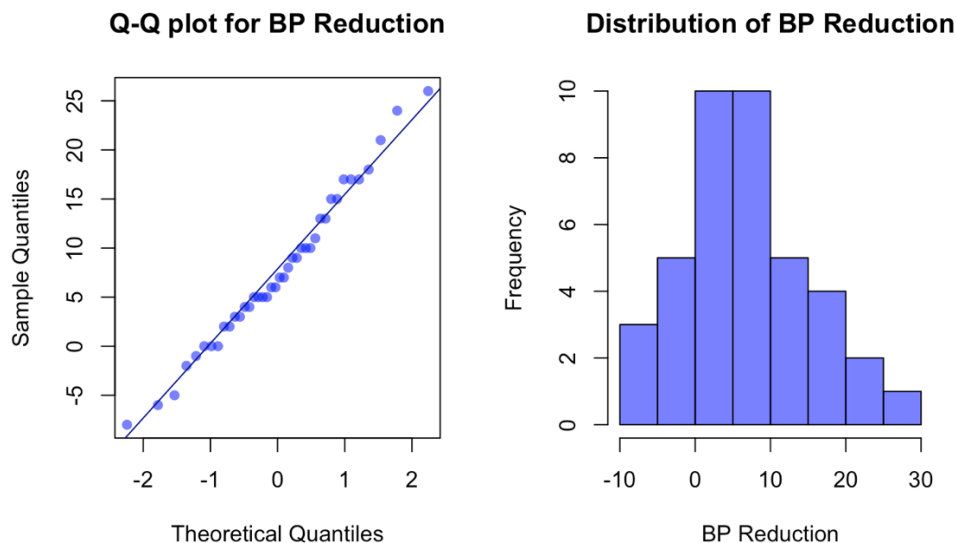
```
##  
## Shapiro-Wilk normality test  
##  
## data: BloodPressure$bp.reduction
```

```
## W = 0.98321, p-value = 0.806
```

Comment: The p-value (0.806) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data is normally distributed.

- **QQ plot and Histogram**

```
par(mfrow=c(1,2))  
# QQ plot  
qqnorm(BloodPressure$bp.reduction, col = rgb(0, 0, 1, 0.6), pch = 16, main =  
"Q-Q plot")  
qqline(BloodPressure$bp.reduction)  
  
# Histogram  
hist(BloodPressure$bp.reduction, xlab = "BP Reduction", ylab = "Frequency", m  
ain = "Distribution of BP Reduction", col = rgb(0, 0, 1, 0.6))
```



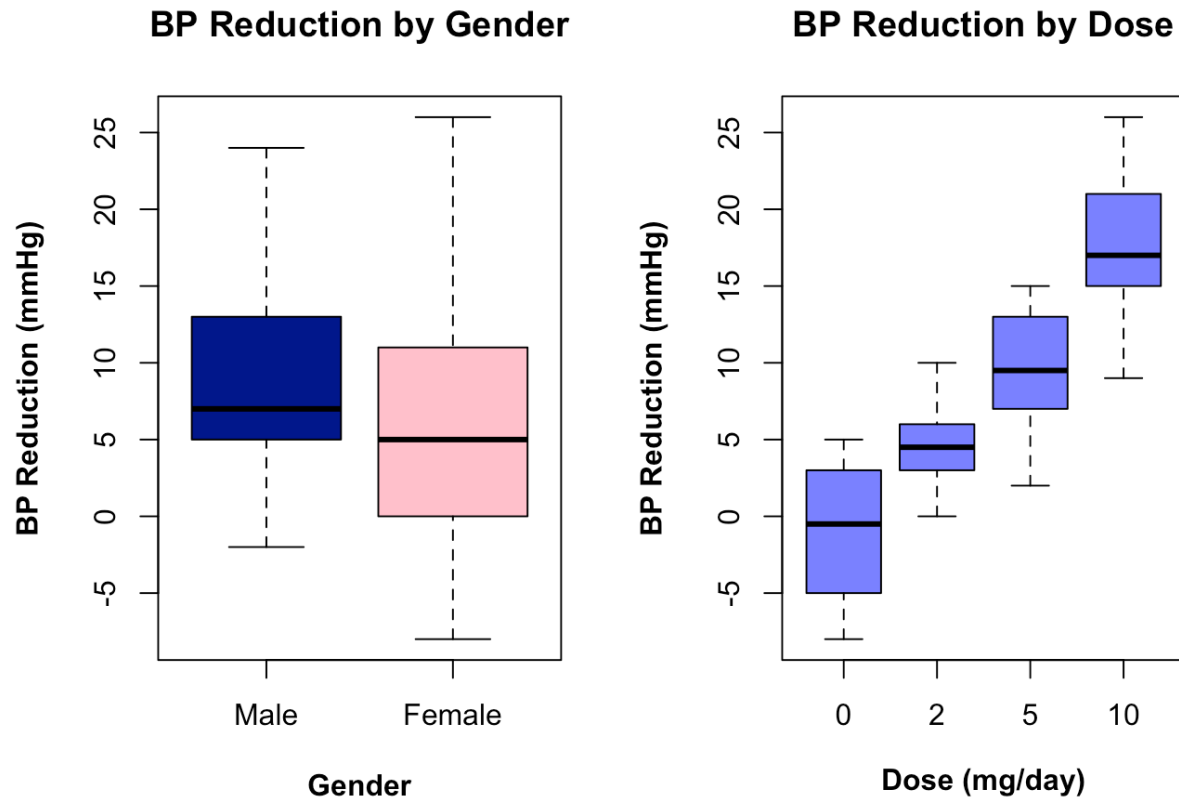
Comment: The QQ plot and Histogram showing the blood pressure reduction data is normally distributed.

.6.2. Homoscedasticity testing:

- **Boxplot**

```
# Homoscedasticity testing  
# Boxplot  
par(mfrow=c(1,2))  
# BP.Reduction vs Gender  
boxplot(bp.reduction~gender, data=BloodPressure, main = "BP Reduction by Gender",  
xlab = expression(bold("Gender")), ylab = expression(bold("BP Reduction  
(mmHg)")), col = c("darkblue", "pink"))  
# BP.Reduction vs Dose
```

```
boxplot(bp.reduction~dose, data=BloodPressure, main = "BP Reduction by Dose",
xlab = expression(bold("Dose (mg/day)")), ylab = expression(bold("BP Reduction (mmHg)")), col = rgb(0, 0, 1, 0.6))
```



- **Levene test**

```
# BP.Reduction vs Gender
```

```
leveneTest(bp.reduction~gender, data=BloodPressure)
```

Output:

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 1  1.8101 0.1865
##      38
```

Comment: The p-value (0.1865) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different genders.

```
# BP.Reduction vs Dose
BloodPressure$dose <- as.factor(BloodPressure$dose)
leveneTest(bp.reduction~dose, data=BloodPressure)
```

Output:

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    3  0.9517  0.426
##           36
```

Comment: The p-value (0.426) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different doses.

- **F-test (var.test)**

```
# BP.Reduction vs Gender
var.test(bp.reduction~gender, data=BloodPressure)
```

Output:

```
##
## F test to compare two variances
##
## data: bp.reduction by gender
## F = 0.51037, num df = 21, denom df = 17, p-value = 0.1447
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.1962979 1.2673945
## sample estimates:
## ratio of variances
##           0.5103705
```

Comment: The p-value (0.1447) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different genders.

F-test can't be applied for BP.Reduction vs Dose, because it is more than 2 groups. Therefore, we applied Barlett test for testing variance between blood pressure reduction and dose.

- **Barlett test**

```
# BP.Reduction vs Dose
bartlett.test(bp.reduction ~ dose, data = BloodPressure)
```

Output:

```
##
##  Bartlett test of homogeneity of variances
##
## data:  bp.reduction by dose
## Bartlett's K-squared = 3.9235, df = 3, p-value = 0.2698
```

Comment: The p-value (0.2698) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different doses.

.7. Statistical inference

```
#Confidence intervals for means of bp.reduction per dose:
BloodPressure %>%
  group_by(dose) %>%
  summarise(
    mean = mean(bp.reduction, na.rm = TRUE),
    ci_90 = qt(0.95, df = n() - 1) * sd(bp.reduction, na.rm = TRUE) / sqrt(n(
  )),
    ci_95 = qt(0.975, df = n() - 1) * sd(bp.reduction, na.rm = TRUE) / sqrt(n(
  )),
    ci_99 = qt(0.995, df = n() - 1) * sd(bp.reduction, na.rm = TRUE) / sqrt(n(
  ))
  )
```

Output:

```
## # A tibble: 4 × 5
##   dose mean ci_90 ci_95 ci_99
##   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     0  -0.9  2.60  3.21  4.61
## 2     2   4.5  1.55  1.92  2.75
## 3     5   9.4  2.17  2.68  3.85
```

```
## 4      10    17.5    3.05    3.76    5.40
```

Comment: Higher doses generally lead to greater blood pressure reduction, suggesting a dose-dependent effect.

dose = 0 group has a CI that includes negative values, indicating no clear reduction.

.8. Hypothesis testing

Hypothesis 1:

We hypothesize that bp.reduction is different between male vs female (in the group that received placebo, i.e. Dose = 0). Assuming normality and homoscedasticity

Set hypothesis setup:

Statistical question: *test whether there is a difference in bp.reduction between males and females in the placebo group (Dose = 0).*

Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$$H_0: \mu_{\text{Male}} = \mu_{\text{Female}}$$

$$H_A: \mu_{\text{Male}} \neq \mu_{\text{Female}}$$

```
# Subset placebo data (dose = 0)
```

```
placebo_data <- BloodPressure[BloodPressure$dose == 0, ]
```

```
# Normality testing
```

```
# Shapiro test for Male data
```

```
shapiro.test(placebo_data$bp.reduction[placebo_data$gender == "Male"])
```

```
##
```

```
##  Shapiro-Wilk normality test
```

```
##
```

```
## data:  placebo_data$bp.reduction[placebo_data$gender == "Male"]
```

```
## W = 0.88482, p-value = 0.3317
```

Comment: The p-value (0.3317) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data for males at placebo dose is normally distributed.

```
# Shapiro test for Female data
```

```
shapiro.test(placebo_data$bp.reduction[placebo_data$gender == "Female"])
```

```
##
```

```
##  Shapiro-Wilk normality test
```

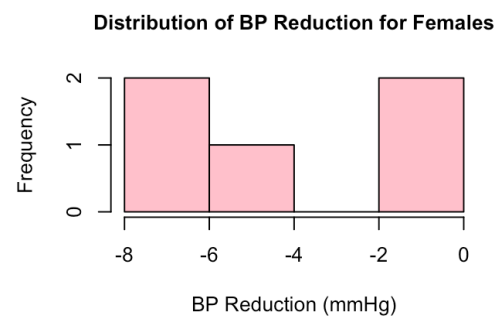
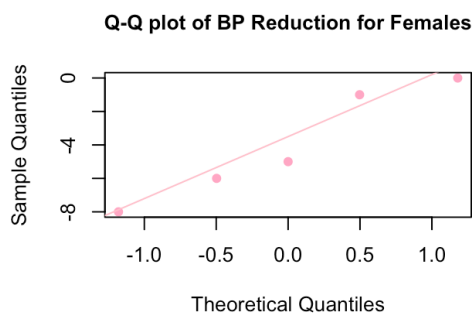
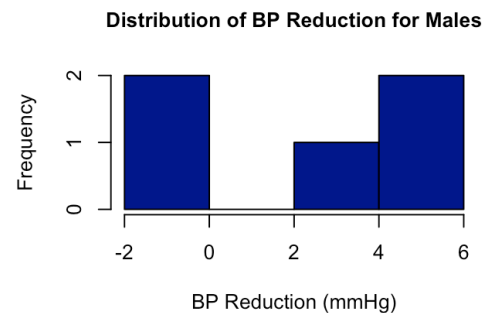
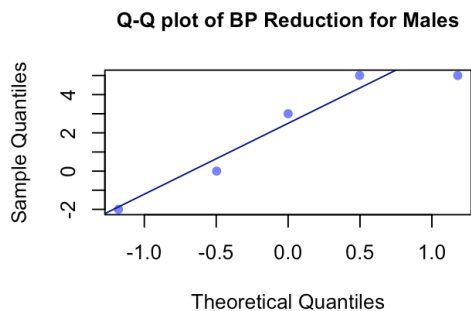
```
##
```

```
## data:  placebo_data$bp.reduction[placebo_data$gender == "Female"]
```

```
## W = 0.92525, p-value = 0.5644
```

Comment: The p-value (0.5644) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data for females at placebo dose is normally distributed.

```
# QQ plot and Histogram
par(mfrow=c(2,2))
# QQ plot for Male data
qqnorm(placebo_data$bp.reduction[placebo_data$gender == "Male"], col = rgb(0,
0, 1, 0.6), pch = 16, cex.main = 1, main = "Q-Q plot of BP Reduction for Male
s")
qqline(placebo_data$bp.reduction[placebo_data$gender == "Male"], col = "darkb
lue")
# Histogram for Male data
hist(placebo_data$bp.reduction[placebo_data$gender == "Male"], xlab = "BP Red
uction (mmHg)", ylab = "Frequency", main = "Distribution of BP Reduction for
Males", col = "darkblue", cex.main = 1)
# QQ plot for Female data
qqnorm(placebo_data$bp.reduction[placebo_data$gender == "Female"], col = rgb(
1, 0.4, 0.6, 0.6), pch = 16, cex.main = 1, main = "Q-Q plot of BP Reduction f
or Females")
qqline(placebo_data$bp.reduction[placebo_data$gender == "Female"], col = "pin
k")
# Histogram for Female data
hist(placebo_data$bp.reduction[placebo_data$gender == "Female"], xlab = "BP R
eduction (mmHg)", ylab = "Frequency", main = "Distribution of BP Reduction fo
r Females", col = "pink", cex.main = 1)
```



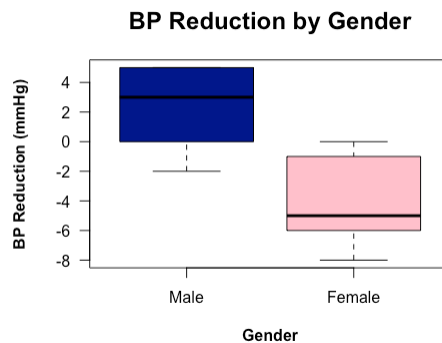
Comment: According to the QQ plots, the blood pressure reduction data for both males females at placebo dose are normally distributed.

```
# Homoscedasticity testing
```

```
par(mfrow=c(1,1))
```

```
# Boxplot
```

```
boxplot(bp.reduction~gender, data=placebo_data, main = "BP Reduction by Gender", xlab = expression(bold("Gender")), ylab = expression(bold("BP Reduction (mmHg)")), col = c("darkblue", "pink"), border = "black", cex.main = 1.5, cex.sub = 1.2, las = 1)
```



```
# Levene test
```

```
leveneTest(bp.reduction~gender, data=placebo_data)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group    1  0.0263 0.8752
##           8
```

Comment: The p-value (0.8752) is more than 0.05. therefore, there is no enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different genders.

```
# F-test (var.test)
```

```
var.test(bp.reduction~gender, data=placebo_data)
```

```
##
## F test to compare two variances
##
## data:  bp.reduction by gender
```

```
## F = 0.84348, num df = 4, denom df = 4, p-value = 0.8729
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.08782088 8.10121216
## sample estimates:
## ratio of variances
##           0.8434783
```

Comment: The p-value (0.8729) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different genders.

```
# Apply Two sample t-test
t.test(bp.reduction ~ gender, data = placebo_data, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: bp.reduction by gender
## t = 3.011, df = 8, p-value = 0.01679
## alternative hypothesis: true difference in means between group Male and group Female is not equal to 0
## 95 percent confidence interval:
##  1.451647 10.948353
## sample estimates:
## mean in group Male mean in group Female
##           2.2           -4.0
```

Comment: The p-value (0.01679) is less than 0.05. therefore, we have enough evidence to reject the null hypothesis in support of alternative hypothesis assuming there is a difference in blood pressure reduction between males and females at placebo dose.

Hypothesis 2:

We hypothesize that bp.reduction is "higher" in the group receiving Dose = 10 compared to the control (Dose = 0). Can you test this hypothesis assuming heteroscedasticity

Set hypothesis setup:

Statistical question: *test whether bp.reduction is higher in the group receiving Dose = 10 compared to the control (Dose = 0).*

Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$$H_0: \mu_{\text{dose}=10} \leq \mu_{\text{dose}=0}$$

$$H_A: \mu_{\text{dose}=10} > \mu_{\text{dose}=0}$$

```
# Subset dose=0 and 10
hypothesis2 <- BloodPressure[BloodPressure$dose == 0 | BloodPressure$dose ==
10, ]

# Normality testing
# Shapiro test for dose=0 data
shapiro.test(hypothesis2$bp.reduction[hypothesis2$dose == 0])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  hypothesis2$bp.reduction[hypothesis2$dose == 0]
## W = 0.94538, p-value = 0.6143
```

Comment: The p-value (0.6143) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data at dose=0 is normally distributed.

```
# Shapiro test for dose=10 data
shapiro.test(hypothesis2$bp.reduction[hypothesis2$dose == 10])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  hypothesis2$bp.reduction[hypothesis2$dose == 10]
## W = 0.95916, p-value = 0.7763
```

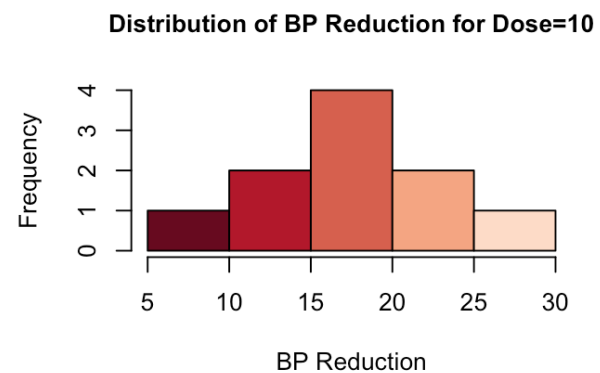
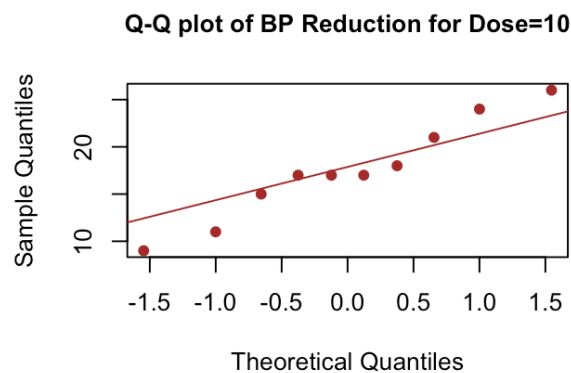
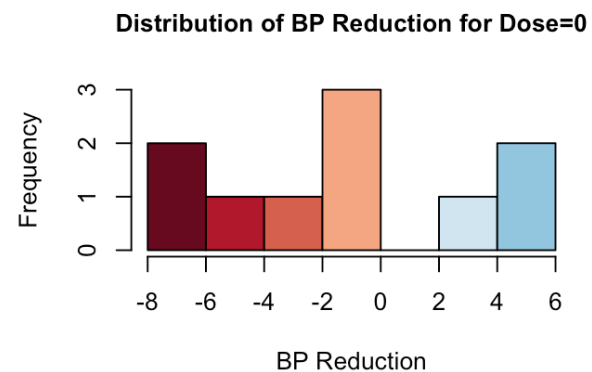
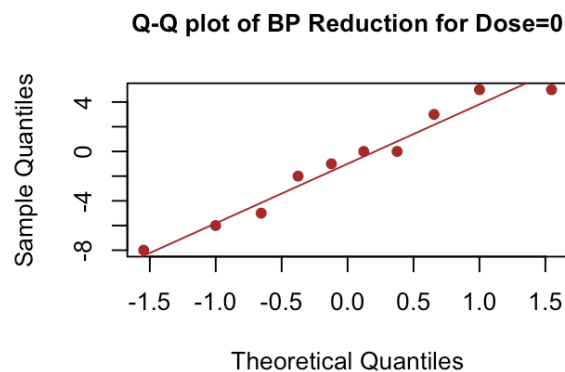
Comment: The p-value (0.7763) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data at dose=10 is normally distributed.

```
# QQ plot and Histogram
par(mfrow=c(2,2))
# QQ plot for dose=0
qqnorm(hypothesis2$bp.reduction[hypothesis2$dose == 0], col = "brown", pch =
16, main = "Q-Q plot of BP Reduction for Dose=0", cex.main = 1)
qqline(hypothesis2$bp.reduction[hypothesis2$dose == 0], col = "brown")
# Histogram for dose=0
hist(hypothesis2$bp.reduction[hypothesis2$dose == 0], xlab = "BP Reduction",
ylab = "Frequency", main = "Distribution of BP Reduction for Dose=0", col = b
```

```

brewer.pal(n = 10, name = "RdBu"), cex.main = 1)
# QQ plot for dose=10
qqnorm(hypothesis2$bp.reduction[hypothesis2$dose == 10], col = "brown", pch =
16, main = "Q-Q plot of BP Reduction for Dose=10", cex.main = 1)
qqline(hypothesis2$bp.reduction[hypothesis2$dose == 10], col = "brown")
# Histogram for dose=10
hist(hypothesis2$bp.reduction[hypothesis2$dose == 10], xlab = "BP Reduction",
ylab = "Frequency", main = "Distribution of BP Reduction for Dose=10", col =
brewer.pal(n = 10, name = "RdBu"), cex.main = 1)

```

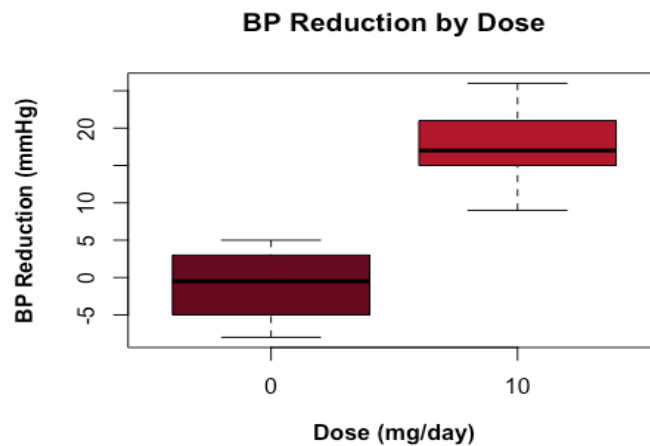


Comment: For dose=0, the QQ plot showed its normal distribution. For dose=10, both QQ plot and histogram showed the data normal distribution.

```

# Homoscedasticity testing
par(mfrow=c(1,1))
# Boxplot
boxplot(bp.reduction~dose, data=hypothesis2, main = "BP Reduction by Dose", x
lab = expression(bold("Dose (mg/day)")), ylab = expression(bold("BP Reduction
(mmHg)")), col = brewer.pal(n = 10, name = "RdBu"))

```



```
# Levene test
```

```
leveneTest(bp.reduction~dose, data=hypothesis2)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 1  0.0207 0.8873
##      18
```

Comment: The p-value (0.8873) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different doses.

```
# F-test (var.test)
```

```
var.test(bp.reduction~dose, data=hypothesis2)
```

```
##
## F test to compare two variances
##
## data:  bp.reduction by dose
## F = 0.72797, num df = 9, denom df = 9, p-value = 0.6439
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.1808169 2.9307941
## sample estimates:
## ratio of variances
```

```
## 0.7279678
```

Comment: The p-value (0.6439) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different doses.

The variance testing showed the variance between doses is homoscedastic while the assumption of the hypothesis assumed the variance is heteroscedastic. According to that, we decided to consider the variance is heteroscedastic and apply Two sample Welch t-test and also Two-Sample t-test.

```
# Apply Two sample Welch t-test
t.test(bp.reduction ~ dose, data = hypothesis2)
```

```
##
## Welch Two Sample t-test
##
## data: bp.reduction by dose
## t = -8.4238, df = 17.565, p-value = 1.403e-07
## alternative hypothesis: true difference in means between group 0 and group
10 is not equal to 0
## 95 percent confidence interval:
## -22.99718 -13.80282
## sample estimates:
## mean in group 0 mean in group 10
## -0.9 17.5
```

Comment: The p-value (1.403e-07) is less than 0.05. therefore, we have enough evidence to reject the null hypothesis in support of alternative hypothesis assuming the blood pressure reduction at dose=10 is higher than dose=0 (control group).

```
# Apply Two sample t-test
t.test(bp.reduction ~ dose, data = hypothesis2, var.equal = TRUE)
```

```
##
## Two Sample t-test
##
## data: bp.reduction by dose
## t = -8.4238, df = 18, p-value = 1.166e-07
```

```
## alternative hypothesis: true difference in means between group 0 and group
10 is not equal to 0

## 95 percent confidence interval:

## -22.98902 -13.81098

## sample estimates:

## mean in group 0 mean in group 10

## -0.9 17.5
```

Comment: The p-value (1.166e-07) is less than 0.05. therefore, we have enough evidence to reject the null hypothesis in support of alternative hypothesis assuming the blood pressure reduction at dose=10 is higher than dose=0 (control group).

Hypothesis 3:

We hypothesis that bp.reduction is different between the different doses (ignoring the gender). Can you perform comparison between the different groups, after assessing the assumptions and performing post-hoc testing.

Set hypothesis setup:

Statistical question: test whether bp.reduction differs between the different dose groups, ignoring gender.

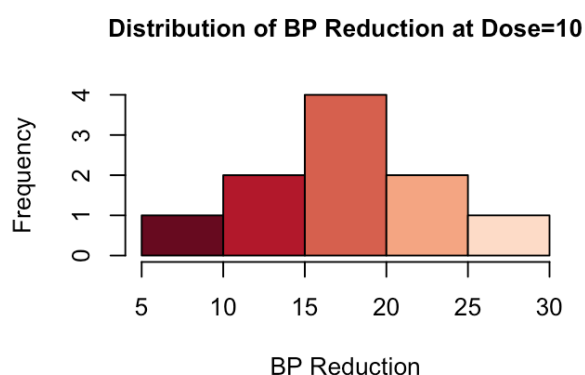
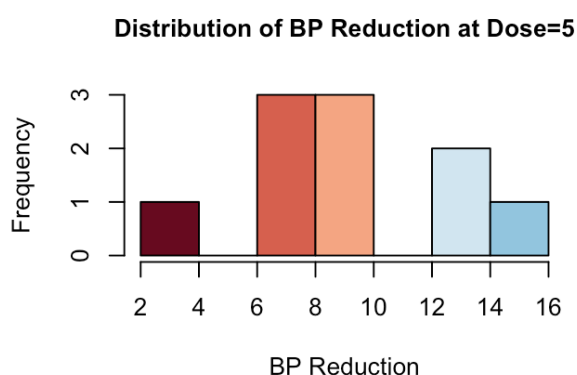
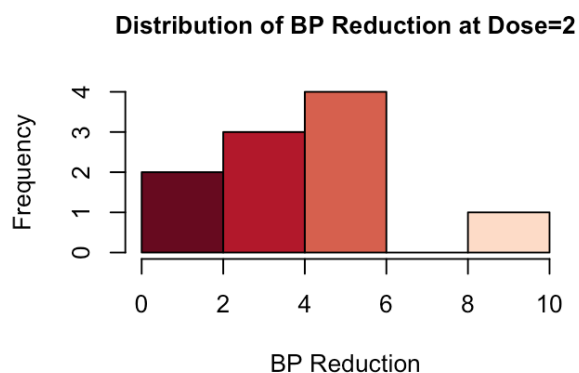
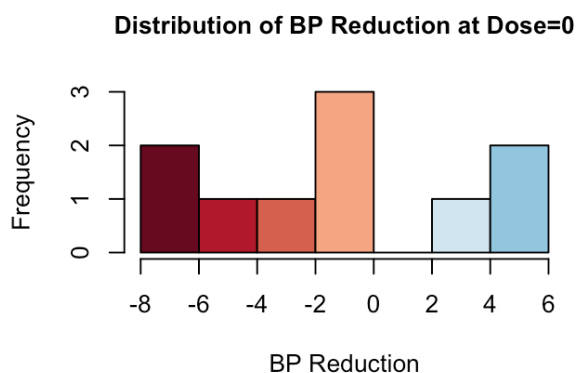
Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$H_0: \mu_{\text{dose}=0} = \mu_{\text{dose}=2} = \mu_{\text{dose}=5} = \mu_{\text{dose}=10}$
 H_A : At least one group mean is different

Verify normality within each group

Histogram

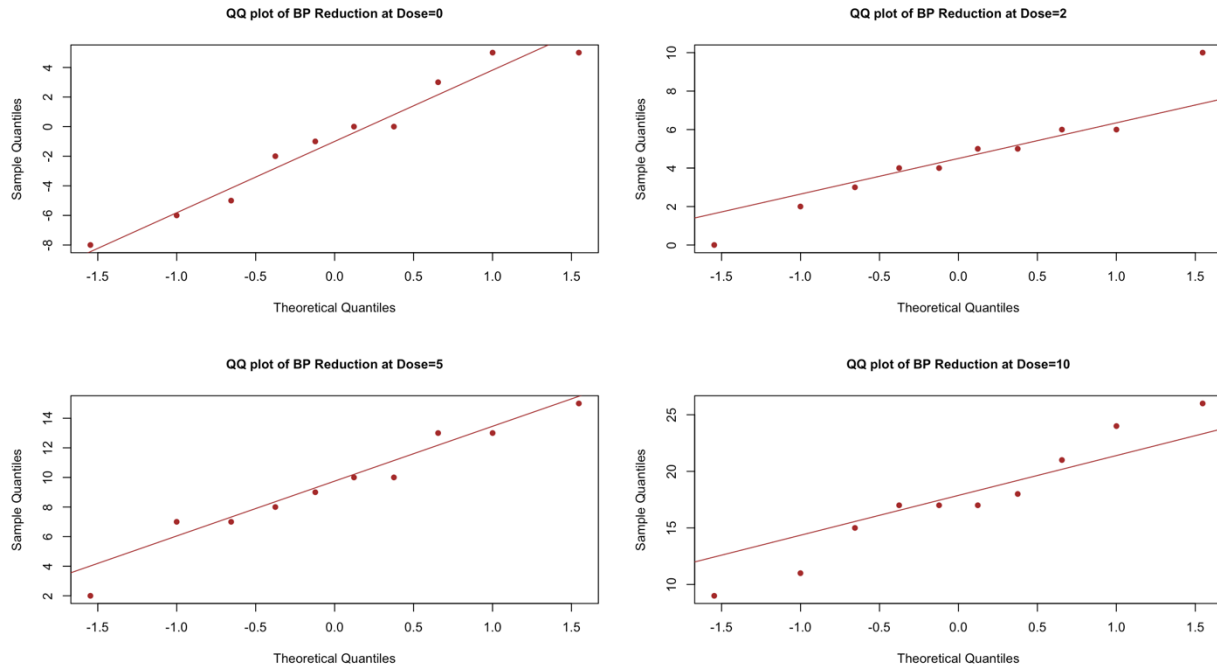
```
par(mfrow=c(2,2))
hist(BloodPressure[BloodPressure$dose==0,]$bp.reduction, xlab = "BP Reduction", ylab = "Frequency", main = "Distribution of BP Reduction at Dose=0", col = brewer.pal(n = 10, name = "RdBu"), cex.main = 1)
hist(BloodPressure[BloodPressure$dose==2,]$bp.reduction, xlab = "BP Reduction", ylab = "Frequency", main = "Distribution of BP Reduction at Dose=2", col = brewer.pal(n = 10, name = "RdBu"), cex.main = 1)
hist(BloodPressure[BloodPressure$dose==5,]$bp.reduction, xlab = "BP Reduction", ylab = "Frequency", main = "Distribution of BP Reduction at Dose=5", col = brewer.pal(n = 10, name = "RdBu"), cex.main = 1)
hist(BloodPressure[BloodPressure$dose==10,]$bp.reduction, xlab = "BP Reduction", ylab = "Frequency", main = "Distribution of BP Reduction at Dose=10", col = brewer.pal(n = 10, name = "RdBu"), cex.main = 1)
```



Comment: For dose=2, 5 and 10, the histogram showed the data normal distribution.

QQ plot

```
qqnorm(BloodPressure$bp.reduction[BloodPressure$dose == 0], col = "brown", pc
h = 16, main = "QQ plot of BP Reduction at Dose=0", cex.main = 1)
qqline(BloodPressure$bp.reduction[BloodPressure$dose == 0], col = "brown")
qqnorm(BloodPressure$bp.reduction[BloodPressure$dose == 2], col = "brown", pc
h = 16, main = "QQ plot of BP Reduction at Dose=2", cex.main = 1)
qqline(BloodPressure$bp.reduction[BloodPressure$dose == 2], col = "brown")
qqnorm(BloodPressure$bp.reduction[BloodPressure$dose == 5], col = "brown", pc
h = 16, main = "QQ plot of BP Reduction at Dose=5", cex.main = 1)
qqline(BloodPressure$bp.reduction[BloodPressure$dose == 5], col = "brown")
qqnorm(BloodPressure$bp.reduction[BloodPressure$dose == 10], col = "brown", p
ch = 16, main = "QQ plot of BP Reduction at Dose=10", cex.main = 1)
qqline(BloodPressure$bp.reduction[BloodPressure$dose == 10], col = "brown")
```



Comment: The QQ plot showed that all doses are normally distributed.

```
# Shapiro test
# For dose=0
shapiro.test(BloodPressure$bp.reduction[BloodPressure$dose == 0])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  BloodPressure$bp.reduction[BloodPressure$dose == 0]
## W = 0.94538, p-value = 0.6143
```

Comment: The p-value (0.6143) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data at dose=0 is normally distributed.

```
# Shapiro test
# For dose=2
shapiro.test(BloodPressure$bp.reduction[BloodPressure$dose == 2])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  BloodPressure$bp.reduction[BloodPressure$dose == 2]
```

```
## W = 0.95757, p-value = 0.7578
```

Comment: The p-value (0.7578) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data at dose=2 is normally distributed.

```
# Shapiro test
# For dose=5
shapiro.test(BloodPressure$bp.reduction[BloodPressure$dose == 5])
```

```
##
## Shapiro-Wilk normality test
##
## data: BloodPressure$bp.reduction[BloodPressure$dose == 5]
## W = 0.95795, p-value = 0.7623
```

Comment: The p-value (0.7623) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data at dose=5 is normally distributed.

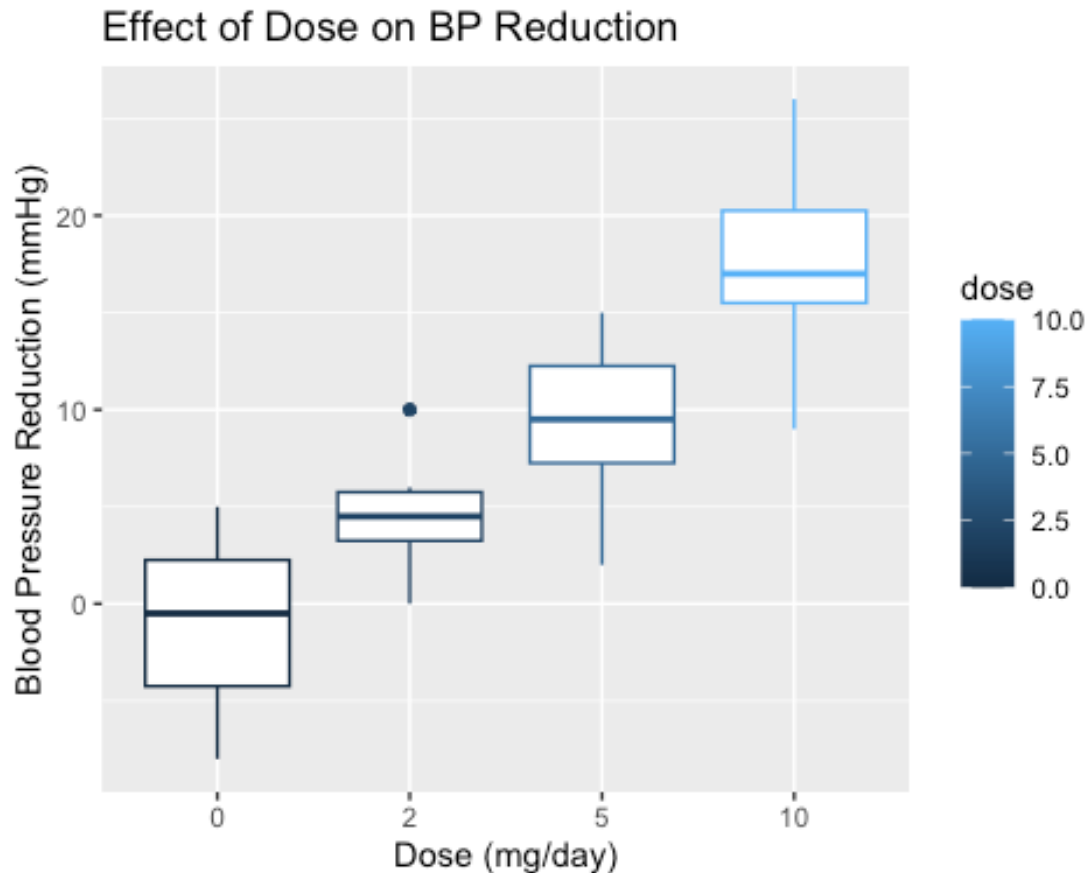
```
# Shapiro test
# For dose=10
shapiro.test(BloodPressure$bp.reduction[BloodPressure$dose == 10])
```

```
##
## Shapiro-Wilk normality test
##
## data: BloodPressure$bp.reduction[BloodPressure$dose == 10]
## W = 0.95916, p-value = 0.7763
```

Comment: The p-value (0.7763) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming blood pressure reduction data at dose=10 is normally distributed.

```
# Verify equality of variances

# Boxplot
par(mfrow=c(1,1))
ggplot(BloodPressure) +
  aes(x = as.factor(dose), y = bp.reduction, color = dose) +
  labs(x = "Dose (mg/day)", y = "Blood Pressure Reduction (mmHg)", title = "Effect of Dose on BP Reduction") +
  geom_boxplot()
```



Levene test

```
leveneTest(bp.reduction~as.factor(dose),data=BloodPressure)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  3  0.9517  0.426
##      36
```

Comment: The p-value (0.426) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction at different doses.

Bartlett test

```
bartlett.test(bp.reduction ~ as.factor(dose), data = BloodPressure)
```

```
##
## Bartlett test of homogeneity of variances
```

```
##
## data:  bp.reduction by as.factor(dose)
## Bartlett's K-squared = 3.9235, df = 3, p-value = 0.2698
```

Comment: The p-value (0.2698) is more than 0.05. therefore, there is not enough evidence to reject the null hypothesis assuming homogeneity of variance between blood pressure reduction and dose.

```
# Apply One-Way ANOVA test
AnovaModel <- aov(bp.reduction ~ as.factor(dose), data = BloodPressure)
summary(AnovaModel)
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## as.factor(dose)  3 1831.1    610.4    35.42 7.71e-11 ***
## Residuals      36   620.3     17.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comment: The p-value (7.71e-11) is less than 0.05. therefore, we have enough evidence to reject the null hypothesis in support of alternative hypothesis assuming there is a difference in blood pressure reduction between different doses (at least one dose is different).

```
#Interpretations of ANOVA results
report(AnovaModel)
```

```
## The ANOVA (formula: bp.reduction ~ as.factor(dose)) suggests that:
##
## - The main effect of as.factor(dose) is statistically significant and large
## (F(3, 36) = 35.42, p < .001; Eta2 = 0.75, 95% CI [0.61, 1.00])
##
## Effect sizes were labelled following Field's (2013) recommendations.
```

```
coef(AnovaModel)
```

```
##      (Intercept)  as.factor(dose)2  as.factor(dose)5  as.factor(dose)10
##           -0.9             5.4             10.3             18.4
```

Posthoc analysis (with Tukey correction)

```
PairComp<-TukeyHSD(AnovaModel)
```

```
PairComp
```

```
##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = bp.reduction ~ as.factor(dose), data = BloodPressure)
##
## $`as.factor(dose)`
##      diff      lwr      upr      p adj
## 2-0     5.4  0.40037278 10.399627 0.0300758
## 5-0    10.3  5.30037278 15.299627 0.0000161
## 10-0   18.4 13.40037278 23.399627 0.0000000
## 5-2     4.9 -0.09962722  9.899627 0.0564941
## 10-2   13.0  8.00037278 17.999627 0.0000002
## 10-5    8.1  3.10037278 13.099627 0.0005742
```

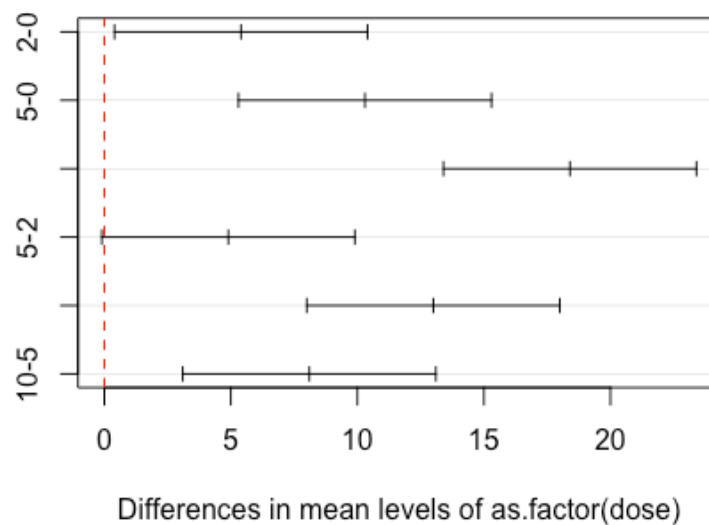
Comment: According to the adjusted p-value of all groups, all groups should have a significant p-value except group (2-5) its adj p-value (0.0564941) is less than 0.05, therefore we don't have enough evidence to reject null hypothesis assuming there is no different between blood pressure reduction values at doses 2 and 5.

```
par(mfrow=c(1,1))
```

```
plot(PairComp)
```

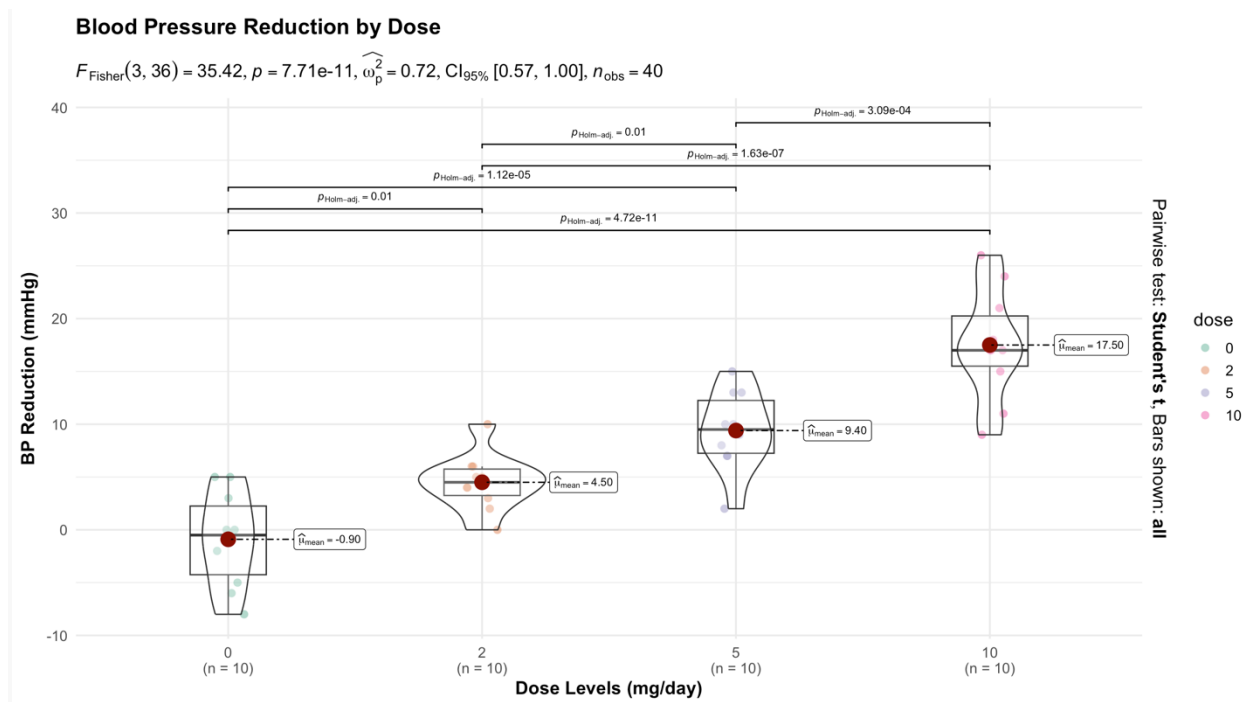
```
abline(v = 0, col = "red", lty = 2)
```

95% family-wise confidence level



#Visualization of ANOVA and post-hoc tests on the same plot

```
ggbetweenstats(  
  data = BloodPressure,  
  x = dose,  
  y = bp.reduction,  
  type = "parametric",  
  var.equal = TRUE,  
  plot.type = "box",  
  pairwise.comparisons = TRUE,  
  pairwise.display = "all",  
  centrality.plotting = TRUE,  
  bf.message = FALSE)+  
  theme_minimal(base_size = 14) +  
  labs(title = expression(bold("Blood Pressure Reduction by Dose")),  
       x = expression(bold("Dose Levels (mg/day)")),  
       y = expression(bold("BP Reduction (mmHg)")))
```



.9. Linear model

Linear model 1:

We hypothesize that bp.reduction is different between male vs female (in the group that received placebo, i.e. Dose = 0). Assuming normality and homoscedasticity

Set model setup:

Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$$H_0: \mu_{\text{Male}} = \mu_{\text{Female}}$$

$$H_A: \mu_{\text{Male}} \neq \mu_{\text{Female}}$$

Set Linear model equation:

$$\text{bp.reduction} = \beta_0 + \beta_1(\text{Gender}) + \epsilon$$

```
# Model 1
# Simple regression model between bp.reduction and gender at dose 0
# Convert gender to numeric values
placebo_data$gender_numeric <- ifelse(placebo_data$gender == "Male", 0, 1)

# Apply regression model
regression1 <- lm(bp.reduction ~ gender_numeric, data = placebo_data)
summary(regression1)
```

```
##
## Call:
```

```
## lm(formula = bp.reduction ~ gender_numeric, data = placebo_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.20  -2.15  -0.10   2.80   4.00
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.200      1.456   1.511  0.1692
## gender_numeric  -6.200      2.059  -3.011  0.0168 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.256 on 8 degrees of freedom
## Multiple R-squared:  0.5312, Adjusted R-squared:  0.4726
## F-statistic: 9.066 on 1 and 8 DF,  p-value: 0.01679
```

```
# Scatter plot for regression1
```

```
plot(placebo_data$gender_numeric, placebo_data$bp.reduction, main = "Gender v
s BP Reduction", xlab = expression(bold("Gender")), ylab = expression(bold("B
P Reduction (mmHg)")), pch = 19, col = rgb(0, 0.5, 0, 0.6), xaxt = "n")
```

```
# Customize gender labels on x-axis
```

```
axis(1, at = c(0, 1), labels = c("Male", "Female"))
```

```
# Regression Line
```

```
abline(regression1, col = "darkgreen", lwd = 2)
```

```
# Confidence Interval for Regression Line
```

```
x_vals <- c(0,1)
```

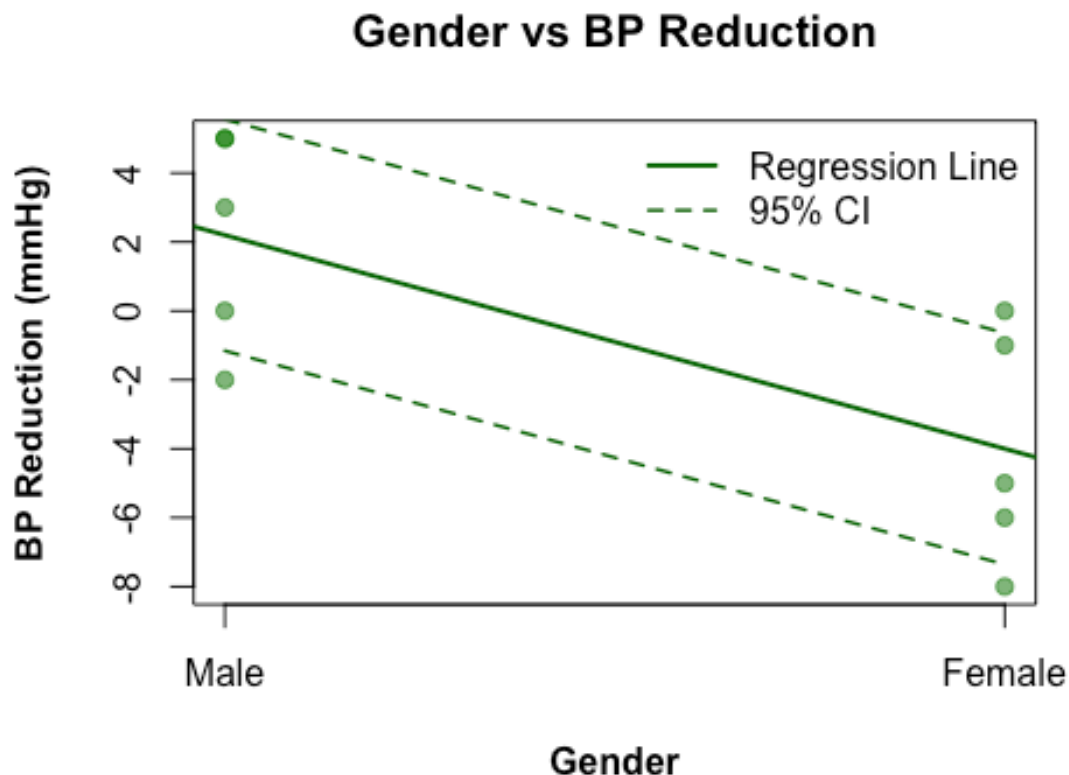
```
predictions <- predict(regression1, newdata = data.frame(gender_numeric = x_v
als), interval = "confidence")
```

```
lines(x_vals, predictions[, "lwr"], col = "darkgreen", lty = 2, lwd = 1.5)
```

```
lines(x_vals, predictions[, "upr"], col = "darkgreen", lty = 2, lwd = 1.5)
```

```
# Legend
```

```
legend("topright", legend = c("Regression Line", "95% CI"),
      col = c("darkgreen", "darkgreen"),
      lty = c(1, 2), lwd = c(2, 1.5),
      bty = "n")
```



Comment: According to the P-value (0.0168) is less than 0.05, the model is reliable. According to the R-squared = 0.5312, the model can capture 53.12% correctly.

Calculate and interpret a 95% confidence interval of the regression slope
`confint(regression1)`

```
##              2.5 %      97.5 %
## (Intercept)  -1.157593  5.557593
## gender_numeric -10.948353 -1.451647
```

Linear model 2:

We hypothesis that bp.reduction is “higher” in the group receiving Dose = 10 compared to the control (Dose =0). Can you test this hypothesis assuming heteroscedasticity

Set model setup:

Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$$H_0: \mu_{\text{dose}=10} \leq \mu_{\text{dose}=0}$$

$$H_A: \mu_{\text{dose}=10} > \mu_{\text{dose}=0}$$

Set Linear model equation:

$$bp.reduction = \beta_0 + \beta_1(Dose) + \epsilon$$

```
# Model 2
# Simple regression model between bp.reduction and doses 0 and 10
# Filter for dose 0 and dose 10
model2_data <- subset(BloodPressure, dose == 0 | dose == 10)
model2_data$dose <- as.numeric(as.character(model2_data$dose))

# Apply regression model
regression2 <- lm(bp.reduction ~ dose, data = model2_data)
summary(regression2)
```

```
##
## Call:
## lm(formula = bp.reduction ~ dose, data = model2_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
##    -8.5    -2.9    -0.3     3.6     8.5
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.9000     1.5445  -0.583   0.567
## dose           1.8400     0.2184   8.424 1.17e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.884 on 18 degrees of freedom
## Multiple R-squared:  0.7977, Adjusted R-squared:  0.7864
## F-statistic: 70.96 on 1 and 18 DF,  p-value: 1.166e-07
```

```
# Scatter plot for regression2
plot(model2_data$dose, model2_data$bp.reduction, main = "Dose vs BP Reduction",
      xlab = expression(bold("Dose (mg/day)")), ylab = expression(bold("BP Reduction (mmHg)")),
      pch = 19, col = rgb(0, 0.5, 0, 0.6))

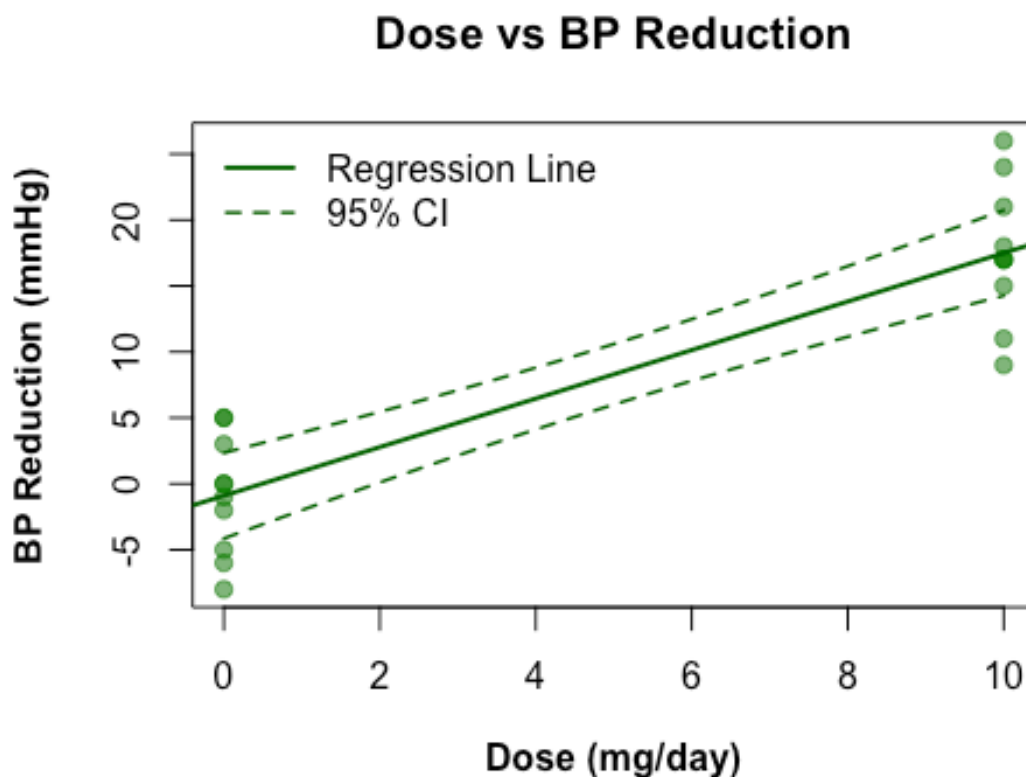
# Regression Line
abline(regression2, col = "darkgreen", lwd = 2)
```

```

# Confidence Interval for Regression Line
x_vals <- seq(min(model2_data$dose), max(model2_data$dose), length.out = 100)
predictions <- predict(regression2, newdata = data.frame(dose = x_vals), interval = "confidence")
lines(x_vals, predictions[, "lwr"], col = "darkgreen", lty = 2, lwd = 1.5)
lines(x_vals, predictions[, "upr"], col = "darkgreen", lty = 2, lwd = 1.5)

# Legend
legend("topleft", legend = c("Regression Line", "95% CI"),
      col = c("darkgreen", "darkgreen"),
      lty = c(1, 2), lwd = c(2, 1.5),
      bty = "n")

```



Comment: According to the P-value ($1.17e-07$) is less than 0.05, the model is reliable. According to the R-squared = 0.7977, the model can capture 79.77% correctly.

```

# Calculate and interpret a 95% confidence interval of the regression slope
confint(regression2)

```

```

##              2.5 %    97.5 %
## (Intercept) -4.144925 2.344925

```

```
## dose          1.381098 2.298902
```

```
# Estimating the average blood pressure reduction for patients that would receive 3mg/day of the medication (bonus).  
predict.lm(regression2, newdata = data.frame(dose=3), level = 0.95)
```

```
##      1  
## 4.62
```

Comment: The predicted value of average blood pressure reduction is 4.62 mmHg.

Linear model 3:

We hypothesize that bp.reduction is different between the different doses (ignoring the gender).

Set model setup:

Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$H_0: \mu_{\text{dose}=0} = \mu_{\text{dose}=2} = \mu_{\text{dose}=5} = \mu_{\text{dose}=10}$

H_A : At least one group mean is different

Set Linear model equation:

$$\text{bp.reduction} = \beta_0 + \beta_1(\text{Dose}) + \epsilon$$

```
# Model 3  
# Simple regression model between bp.reduction and dose  
  
# Apply regression model  
BloodPressure$dose <- as.numeric(as.character(BloodPressure$dose))  
regression3 <- lm(bp.reduction ~ dose, data = BloodPressure)  
summary(regression3)
```

```
##  
## Call:  
## lm(formula = bp.reduction ~ dose, data = BloodPressure)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -8.8965 -1.9648  0.0011  2.5374  8.1035   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  0.03304    0.97868    0.034    0.973
## dose        1.78634    0.17234   10.366 1.25e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.105 on 38 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7319
## F-statistic: 107.4 on 1 and 38 DF,  p-value: 1.247e-12
```

```
# Scatter plot for regression3
```

```
plot(BloodPressure$dose, BloodPressure$bp.reduction, main = "Dose vs BP Reduc
tion", xlab = expression(bold("Dose (mg/day)")), ylab = expression(bold("BP R
eduction (mmHg)")), pch = 19, col = rgb(0, 0.5, 0, 0.6))
```

```
# Regression Line
```

```
abline(regression3, col = "darkgreen", lwd = 2)
```

```
# Confidence Interval for Regression Line
```

```
x_vals <- seq(min(BloodPressure$dose), max(BloodPressure$dose), length.out =
100)
```

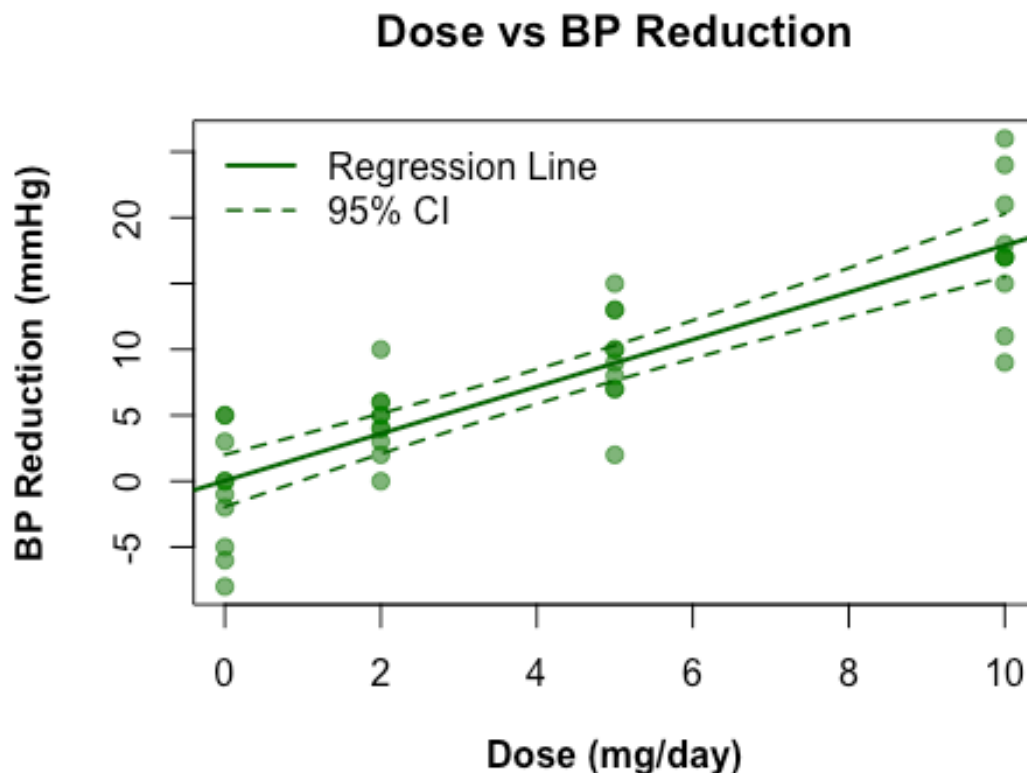
```
predictions <- predict(regression3, newdata = data.frame(dose = x_vals), inte
rval = "confidence")
```

```
lines(x_vals, predictions[, "lwr"], col = "darkgreen", lty = 2, lwd = 1.5)
```

```
lines(x_vals, predictions[, "upr"], col = "darkgreen", lty = 2, lwd = 1.5)
```

```
# Legend
```

```
legend("topleft", legend = c("Regression Line", "95% CI"),
      col = c("darkgreen", "darkgreen"),
      lty = c(1, 2), lwd = c(2, 1.5),
      bty = "n")
```



Comment: According to the P-value ($1.25e-12$) is less than 0.05, the model is reliable. According to the R-squared = 0.7387, the model can capture 73.87 % correctly.

```
# Calculate and interpret a 95% confidence interval of the regression slope
confint(regression3)
```

```
##              2.5 %    97.5 %
## (Intercept) -1.948188 2.014267
## dose         1.437469 2.135218
```

```
# Estimating the average blood pressure reduction for patients that would receive 3mg/day of the medication (bonus).
```

```
predict.lm(regression3, newdata = data.frame(dose=3), level = 0.95)
```

```
##          1
## 5.39207
```

Comment: The predicted value of average blood pressure reduction is 5.39207 mmHg.

Linear model 4 (extra):

We hypothesis that *bp.reduction* is different between the different doses (including the gender).

Set model setup:

Set Null Hypothesis (H_0) and Alternative Hypothesis (H_A):

$$H_0: \mu_{\text{dose}=0} = \mu_{\text{dose}=2} = \mu_{\text{dose}=5} = \mu_{\text{dose}=10}$$

H_A : At least one group mean is different

Set Linear model equation:

$$\text{bp.reduction} = \beta_0 + \beta_1(\text{Dose}) + \beta_2(\text{Gender}) + \epsilon$$

```
# Model 4 (extra)
# Multiple regression model between bp.reduction and dose + gender

# Apply regression model
plot(data.frame(BloodPressure$bp.reduction, BloodPressure$dose, BloodPressure
$gender), col = rgb(0, 0.5, 0, 0.6), pch = 16, cex = 1.5, main = "Pairwise Re
lationships")
```



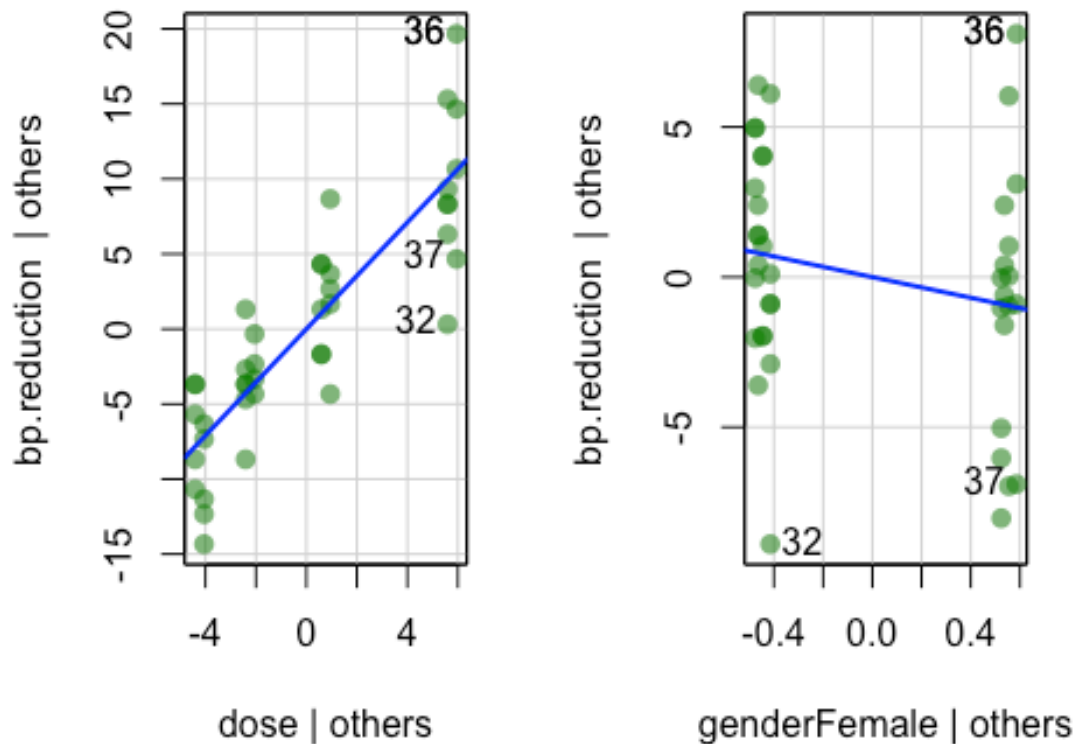
```
regression4 <- lm(bp.reduction~dose+gender, data = BloodPressure)
summary(regression4)
```

```
##
## Call:
## lm(formula = bp.reduction ~ dose + gender, data = BloodPressure)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6098 -2.7311  0.1899  2.4279  9.1109
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.8525     1.1480   0.743   0.462
## dose           1.7757     0.1708  10.397 1.57e-12 ***
## genderFemale  -1.7207     1.2932  -1.331   0.191
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.064 on 37 degrees of freedom
## Multiple R-squared:  0.7507, Adjusted R-squared:  0.7372
## F-statistic: 55.7 on 2 and 37 DF,  p-value: 6.928e-12
```

*Comment: According to the P-value (6.928e-12) is less than 0.05, the model is reliable.
According to the adjusted R-squared = 0.7372, the model can capture 73.72 % correctly.*

```
# Produce added variable plot
avPlots(regression4, col = rgb(0, 0.5, 0, 0.6), pch = 16, cex = 1.2, grid = T
RUE, main = "Added Variable Plots")
```

Added Variable Plots



Calculate and interpret a 95% confidence interval of the regression slope
`confint(regression4)`

```
##           2.5 %    97.5 %
## (Intercept) -1.473704 3.1786181
## dose        1.429657 2.1218062
## genderFemale -4.340907 0.8995045
```

Estimating the average blood pressure reduction for patients that would receive 3mg/day of the medication (bonus).
`predict.lm(regression4, newdata = data.frame(dose=3, gender="Male"), level = 0.95)`

```
##           1
## 6.179651
```

Comment: The predicted value of average blood pressure reduction is 6.179651 mmHg.

```
predict.lm(regression4, newdata = data.frame(dose=3, gender="Female"), level  
= 0.95)
```

```
##          1  
## 4.45895
```

Comment: The predicted value of average blood pressure reduction is 4.45895 mmHg.